

ATMA FATHONI MOZA

Electronics Engineering Technology Student | Hardware Development | Embedded Systems | IoT

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PROFESSIONAL PROFILE

Electronics Engineering Technology student at Politeknik Negeri Batam with a current GPA of 3.71/4.00 and hands-on experience in hardware development, industrial maintenance, embedded systems, PCB design and fabrication, IoT, CAD, and 3D-print prototyping. Currently supports hardware integration, component qualification, technical documentation, and mechanical validation during a Hardware Development internship. Previous industrial experience includes preventive maintenance, machine calibration, troubleshooting, and supervised PLC ladder-logic work. Comfortable working across electronics, firmware, mechanical design, testing, and documentation in team environments.

WORK EXPERIENCE

Hardware Development Intern - PT Kelola Insan Digital

Development Division | Aug 2026-Present | Scheduled through Jan 2027

- Support prototyping and integration of ESP32 and ESP32-S3 IoT hardware through circuit assembly, wiring validation, functional testing, and technical documentation.
- Qualify components by checking variants, pinouts, connectors, physical dimensions, electrical compatibility, soldering quality, and known-good operation.
- Prepare and update bills of materials, component records, wiring guides, and Fritzing assets for repeatable assembly.
- Support 3D-printable part validation through STEP/STL handling, clearance and fit checks, and Bambu Lab print iterations.

Electrical Technician Apprentice - PT Cladtek Bi-Metal Manufacturing

Maintenance Department | Feb-May 2021

- Assisted with preventive maintenance, machine calibration, and troubleshooting of industrial electrical and electronic equipment.
- Supported control-panel and welding-machine inspection using multimeters and basic diagnostic methods.
- Assisted with PLC ladder-logic programming and functional checks under technician supervision.
- Performed testing, soldering, and maintenance activities while following workplace safety procedures.

EDUCATION

Politeknik Negeri Batam

Bachelor of Applied Engineering in Electronics Engineering Technology | 2023-Present | GPA 3.71/4.00 through Semester 7

SMK Negeri 1 Batam

Industrial Electronics Engineering | 2019-2022

SELECTED PROJECTS

GESTURA Portable Communication Device | GitHub

Team Project | 2026-Present | Personal role: Hardware Integration and 3D-Print Enclosure Design

Tools: Raspberry Pi 4 | Webcam | Display | Microphone | Speakers | Amplifier | 3D Printing

- Assembled and integrated the Raspberry Pi 4 hardware, including the webcam, display, microphone, speakers, and amplifier.
- Designed the 3D-printable enclosure and supported physical placement, wiring, and fit of the system components.
- The team system supported 125 SIBI/BISINDO gestures plus IDLE; real-device testing achieved 66.24% accuracy (414/625 attempts) with a 3.91-second average response time.

IoT Smart Prayer Room | GitHub

Team Project | Personal role: Enclosure Design and System-Flow Planning

Tools: Raspberry Pi | Python | Webcam | Relays | Pan-Tilt Servos | Web Monitoring

- Designed the enclosure and planned the system workflow for a Raspberry Pi-based room automation prototype.
- The team system used webcam-based human detection to switch lights and a fan through relays according to room occupancy.
- Completed a functional prototype with web monitoring and a two-axis pan-tilt camera.

IoT pH Monitoring System | GitHub

Team Project | Personal role: Hardware Assembly and ESP32 Programming

Tools: ESP32 | Arduino IDE/C++ | 4502C Module | E-201 Electrode | Local Display | ThingSpeak

- Assembled the hardware and programmed the ESP32 monitoring system using Arduino IDE and C++.
- Integrated the 4502C module, E-201 electrode, local display, and ThingSpeak remote monitoring.
- Calibration testing achieved a measurement error below 5%.

CORE COMPETENCIES

Hardware and integration: Electronic prototyping, circuit assembly, wiring, sensor and actuator integration, component qualification.

Embedded systems: Raspberry Pi 4, ESP32, ESP32-S3, Arduino Nano, Arduino IDE, C++, Python.

PCB and mechanical design: Altium Designer, PCB layout and fabrication, SolidWorks, AutoCAD, enclosure and chassis design, STEP/STL handling.

Testing and maintenance: Multimeter testing, calibration, troubleshooting, preventive maintenance, control-panel inspection, functional validation.

IoT and control: ThingSpeak, web monitoring, relay logic, PLC ladder logic, basic PLC concepts.

Technical documentation: Bills of materials, component records, wiring guides, test documentation, and project README files.

SELECTED PROJECTS CONTINUED

Trainer Kit PLC Model Workbench | GitHub

Team Project | Pre-PLC Relay and Push-Button Logic Stage

Tools: Workbench Frame | Relay Box | Push-Button Box | Wiring | Functional Testing

- Helped construct the workbench frame, relay box, and push-button input box from the initial fabrication stage.
- Assembled and wired the push-button input and relay-output logic used before planned PLC integration.
- Completed functional testing of the physical workbench and its relay-based input-output stage.

Line-Following Robot | GitHub

Team Project | Personal role: PCB, Chassis Parts, and Arduino Programming

Tools: Arduino Nano | Arduino IDE/C++ | Sensor PCB | Motor-Driver PCB | Chassis Design

- Designed the sensor and motor-driver PCB layouts and fabricated the completed circuit boards.
- Designed chassis parts and programmed the Arduino Nano using Arduino IDE and C++.
- Integrated and tested the system until the robot successfully followed the test track.

SOFTWARE AND TOOLS

Altium Designer: PCB schematic capture, sensor-board layout, and motor-driver layout.
SolidWorks and AutoCAD: Enclosure parts, chassis parts, and technical drawings.
Arduino IDE and C++: ESP32 and Arduino Nano programming, testing, and calibration routines.
Python: General scripting and Raspberry Pi project support.
Fritzing: Component assets, wiring references, and build documentation.
Bambu Lab workflow: 3D-print preparation, print iteration, and fit checking.
ThingSpeak and GitHub: IoT monitoring, project documentation, and basic version control.

CERTIFICATIONS AND TRAINING

Chinese and Vocational Skills Training Program in Boiler and Steam Turbine Operations

Changjiang Institute of Technology | Hosted at Politeknik Negeri Batam | 26 Jan-6 Feb 2026

Industrial Electronics Competency Certification

PT Sumitomo Wiring Systems | 2022

PROFESSIONAL STRENGTHS

Team collaboration, structured troubleshooting, attention to detail, technical documentation, adaptability across hardware and software tasks, and industrial safety awareness.

LANGUAGES AND PORTFOLIO

Indonesian - Native | **English** - Intermediate conversational proficiency

GitHub Profile | **LinkedIn Profile**